
Chapter 4

Bayes Classifier

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Outline

- ❑ Introduction
- ❑ The naïve Bayes Probabilistic model
- ❑ Constructing a Classifier from the probability model.
- ❑ Naïve Bayes Classifier for continuous data.

1. Introduction to Naïve Bayes Classifier

- A **naïve Bayes classifier** is a simple probabilistic classifier based on applying Bayes theorem where every feature is assumed to be *class-conditionally independent*.
- Naïve Bayes classifiers assume that the effect of a feature value on a given class is *independent of* the values of other features. This assumption is called **class-conditionally independence**. It is made to simplify the computation and in this sense, it is considered to be *naïve*.
- The assumption is fairly strong and sometimes is not applicable.
- But studies comparing classification algorithms have found the naïve Bayes classifier to be comparable in performance with decision trees and neural network classifiers.
- Bayes classifiers also exhibited *high accuracy* and *speed* when applied to large databases.

2. The Naïve Bayes Probabilistic Model

- The probabilistic model for a classifier is a conditional model

$$P(C|F_1, \dots, F_n)$$

where n is the number of features.

- Over a class variable C with a small number of classes, conditional on several feature variables F_1 through F_n .
- Using Bayes theorem, we can write
$$P(C|F_1, \dots, F_n) = P(F_1, \dots, F_n|C)P(C)/P(F_1, \dots, F_n).$$
- The denominator does not depend on C and the values of the features F_i are given, so the denominator is effectively constant. The numerator is equivalent to the joint probability model

$$\begin{aligned} &P(C, F_1, \dots, F_n) \\ \Rightarrow &P(C|F_1, \dots, F_n) \propto P(C, F_1, \dots, F_n) \end{aligned}$$

Note: $P(C|A) = P(A|C).P(C)/P(A)$

$$P(C \cap A) = P(C|A).P(A)$$

Which can be rewritten as follows, using repeated applications of the definition of conditional probability:

$$\begin{aligned}P(C, F_1, \dots, F_n) &= P(C)P(F_1, \dots, F_n|C) \\&= P(C)P(F_1|C)P(F_2, \dots, F_n|C, F_1) \\&= P(C)P(F_1|C)P(F_2|C, F_1)P(F_3, \dots, F_n|C, F_1, F_2) \\&= P(C)P(F_1|C)P(F_2|C, F_1)P(F_3|C, F_1, F_2)P(F_4, \dots, F_n|C, F_1, F_2, F_3)\end{aligned}$$

- And so forth. Using the conditional independent assumption, each feature F_i is conditionally independent of every other feature F_j for $j \neq i$. This means that

$$P(F_i|C, F_j) = P(F_i|C)$$

- And so the joint model can be expressed as

$$\begin{aligned}P(C, F_1, \dots, F_n) &= P(C)P(F_1|C)P(F_2|C)P(F_3|C)\dots P(F_n|C) \\&= P(C)\prod_{i=1}^n P(F_i | C)\end{aligned}\tag{4.1}$$

Models of this form are much more manageable, since they use prior probabilities of classes $P(C)$ and independent probability distributions $P(F_i|C)$.

Parameter estimation

- The training set contains examples belonging to different classes. The ratio of the number of examples of a particular class and the total number of examples gives the *class prior probability* for that particular class. In this way, the *class prior probability* can be calculated for every class.
- Consider the training set:
Total number of examples = 100
Number of examples of Class 1 = 40
Number of examples of Class 2 = 30
Number of examples of Class 3 = 30
- Therefore,
Prior probability of Class 1 = $40/100 = 0.4$
Prior probability of Class 2 = $30/100 = 0.3$
Prior probability of Class 3 = $30/100 = 0.3$
- Out of the 40 examples of Class 1, if a binary feature takes 0 in 30 examples and 1 in 10 examples, then probability that this feature is 0 in this class will be $30/40 = 0.75$.

3. Constructing a Classifier from the probability model

- The naïve Bayes classifier combines the *Bayes probability model* with a *decision rule*. One common rule is to pick the hypothesis that is most probable; this is known as the *maximum a posterior* or *MAP decision rule*. The corresponding classifier is the function “classify” defined as follows:

$$\text{classify}(f_1, \dots, f_n) = \arg \max_c P(C = c) \prod_{i=1}^n P(F_i = f_i \mid C = c) \quad (4.2)$$

Given a tuple $(f_1, \dots, f_n)$, it will be assigned to the class having the highest posterior probability.

Example

- Suppose we have a data set with 10 patterns as follows

Cook	Mood	Cuisine	Tasty

Sita	Bad	Indian	Yes
Sita	Good	Continental	Yes
Asha	Bad	Indian	No
Asha	Good	Indian	Yes
Usha	Bad	Indian	Yes
Usha	Bad	Continental	No
Asha	Bad	Continental	No
Asha	Good	Continental	Yes
Usha	Good	Indian	Yes
Usha	Good	Continental	No

Consider a new pattern: *Cook = Sita, Mood = Bad, Cuisine = Continental*.
We need to classify this example as *Tasty = yes* or *Tasty = no*.

- Since there are 6 examples out of 10 with $Tasty = yes$, the prior probability $P(Tasty = yes)$ is $6/10 = 0.6$. The prior probability $P(Tasty = no)$ is $4/10 = 0.4$.
- There are 2 examples with $Cook = Sita$ and $Tasty = yes$ and 0 example with $Cook = Sita$ and $Tasty = no$.
- The probability for $Cook = Sita$ given that $Tasty = yes$ is
 $P(Cook = Sita \mid Tasty = yes) = 2/6 = 0.33$
 $P(Cook = Sita \mid Tasty = no) = 0$
- The probability is zero, and since it is multiplied by other probabilities, those probabilities will also be zero. To avoid this, a small value is taken. Let it be 0.01.
- There are 2 examples with $Mood = Bad$ and $Tasty = yes$ and 3 examples with $Mood = Bad$ and $Tasty = no$. Therefore
 $P(Mood = Bad \mid Tasty = yes) = 2/6 = 0.33$
 $P(Mood = Bad \mid Tasty = no) = 3/4 = 0.75$

Example (cont.)

- There are 2 examples with *Cuisine* = *Continental* and *Tasty* = yes and 3 examples with *Cuisine* = *Continental* and *Tasty* = no. Therefore

$$P(\text{Cuisine} = \text{Continental} \mid \text{Tasty} = \text{yes}) = 2/6 = 0.33$$

$$P(\text{Cuisine} = \text{Continental} \mid \text{Tasty} = \text{no}) = 3/4 = 0.75$$

- Therefore

$$P(\text{Tasty} = \text{yes} \mid X) = P(\text{Tasty} = \text{yes}).P(\text{Cook}=\text{Sita} \mid \text{Tasty}=\text{yes}).$$

$$P(\text{Mood}=\text{bad} \mid \text{Tasty}=\text{yes}).P(\text{Cuisine}=\text{continental} \mid \text{Tasty}=\text{yes}) \\ = 0.6 \times 0.33 \times 0.33 \times 0.33 = 0.0216$$

$$P(\text{Tasty} = \text{no} \mid X) = P(\text{Tasty} = \text{no}).P(\text{Cook}=\text{Sita} \mid \text{Tasty}=\text{no}).$$

$$P(\text{Mood}=\text{bad} \mid \text{Tasty}=\text{no}).P(\text{Cuisine}=\text{continental} \mid \text{Tasty}=\text{no}) \\ = 0.4 \times 0.01 \times 0.75 \times 0.75 = 0.00225$$

- The new pattern is classified as belonging to the class *Tasty* = yes as $P(\text{Tasty} = \text{yes} \mid X) > P(\text{Tasty} = \text{no} \mid X)$.

4. Naïve Bayes Classifier for continuous data

- If F_i is continuous-valued attribute, then we need to do a bit more work, but the calculation is pretty straightforward. A continuous-valued attribute is typically assumed to have a *Gaussian distribution* with a mean μ and standard deviation σ , defined by:

$$g(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

so that

$$P(F_i = f_i | C = c) = g(f_i, \mu_c, \sigma_c)$$

where μ_c and σ_c are the mean and standard deviation, respectively, of the values of attribute F_i for training tuples of class c .

We then plug these two quantities into Eq. (4.2), together with f_i , in order to estimate $P(F_i = f_i | C = c)$.

Example

- For example, let $\mathbf{X} = (35, \$40,000)$, where F_1 and F_2 are the attributes *age* and *income*, respectively. Let the class label attribute be *buys_computer*. The associated class label for $\mathbf{X}$ is *yes* (i.e., *buys_computer* = *yes*). Let's suppose that *age* is a continuous-valued attribute. Suppose that from the training set, we find that customers in dataset D who buy a computer are 38 ± 12 years of age. In other words, for attribute *age* and this class, we have $\mu = 38$ years and $\sigma = 12$. We can plug these quantities, along with $F_1 = 35$ for our tuple $\mathbf{X}$, into Eq. (4.2) to estimate $P(\text{age} = 35 | \text{buys_computer} = \text{yes})$.
- Naïve Bayes classifier with Gaussian distribution is called ***Gaussian Naïve Bayes classifier***.

$$\frac{1}{\sqrt{2\pi \cdot 12}} e^{-\frac{(35-38)^2}{2 \cdot 12^2}}$$

Compare Bayesian Classifier with NNC

- When classifying a new pattern using the *nearest neighbor classifier*, we have to find the distance from the test pattern to every pattern in every class and classify it as belonging to the class of the nearest pattern.
- In the *naïve Bayesian classifier*, the *posterior probability* of the new pattern has to be computed for every class.
- Usually, the number of classes $C \ll n$ (n : the number of training patterns), therefore, the Naïve Bayes classifier is much faster.
- There is no design time for NNC whereas for the Naïve Bayes classifier, it is necessary to calculate the probability for each class using the training data.
- Note: Naïve Bayesian Classifier is one of top ten algorithms in Data Mining.

Tools for Naïve Bayesian Classifier

- Naïve Bayesian Classifier is available in:
 - ❑ MATLAB
 - ❑ The data mining software Weka
 - ❑ RapidMiner
 - ❑ Scikit-learn

Appendix

- Probability
- Axioms of probability
- Conditional probability

Probability

Prior probability: probability in the absence of any other information

$P(A)$: probability of event A

Example:

$$P(\text{Dice} = 2) = 1/6$$

random variable: *Dice*

domain = $\langle 1, 2, 3, 4, 5, 6 \rangle$

probability distribution: $P(\text{Dice}) = \langle 1/6, 1/6, 1/6, 1/6, 1/6, 1/6 \rangle$

Axioms of probability

- $0 \leq P(A) \leq 1$
- $P(\text{true}) = 1$ and $P(\text{false}) = 0$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Derived properties:

- $P(\neg A) = 1 - P(A)$
- $P(U) = P(A_1) + P(A_2) + \dots + P(A_n)$

$U = A_1 \cup A_2 \cup \dots \cup A_n$ collectively exhaustive

$A_i \cap A_j = \text{false}$ mutually exclusive (if they can not
occur at the same time)

Events A and B are *independent* if

$$P(A \cap B) = P(A).P(B)$$

Note: $P(A \cap B)$ (joint probability) can be written as $P(A,B)$

Conditional probability

Conditional probability: probability in the presence of some evidence.

$P(E|F)$ is the probability of the occurrence of event E given that F occurred

$$P(\text{Dice} = 2 \mid \text{Dice is even}) = 1/3$$

$$P(\text{Dice} = 2 \mid \text{Dice is odd}) = 0$$

Bayes' formula

$$P(A \mid B) = P(A \cap B)/P(B)$$

$$P(A \cap B) = P(A \mid B).P(B)$$

Because $\cap$ is commutative, we have:

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A) \quad \text{which gives us Bayes' formula}$$

$$P(B|A) = P(A|B)P(B)/P(A)$$

Example (Conditional Probability)

Example:

S = stiff neck

M = meningitis //bệnh viêm màng não

$$P(S | M) = 0.5$$

$$P(M) = 1/50000$$

$$P(S) = 1/20$$

$$P(M | S) = P(S | M) \cdot P(M) / P(S) = 0.5 \cdot (1/50000) \cdot (1/20) \\ = 1/5000$$

References

- M. N. Murty and V. S. Devi, 2011, *Pattern Recognition – An Algorithmic Approach*, Springer.
- S. L. Ting, W. H. Ip, A. H. C. Tsang, *Is Naïve Bayes a Good Classifier for Document Classification?*, Int. Journal of Software Engineering and Its Applications, Vol. 5, No. 3, July, 2011.

Terminology

- prior probability: *tiền xác suất*, conditional probability: *xác suất có điều kiện (hậu xác suất)*, $P(A|B)$, $P(A \cap B)$, $P(A \cup B)$, mutually exclusive: *loại trừ lẫn nhau*, collectively exhaustive: *vét cạn toàn cục*, Bayesian formula: *công thức Bayes*, Bayesian theorem: *định lý Bayes*, Naïve Bayes classifier: *bộ phân lớp Bayes đơn giản*, class-conditionally independence: *độc lập có điều kiện về lớp*, maximum a posterior decision rule: *luật quyết định hậu xác suất cực đại*.